LOCALIZATION OF THE CRYSTALLINE SITE BY OPEN IMMERSIONS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file crystalline.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 07MF: sources/stacks-chow/tags.txt, source lines 9725–9725. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04H1: sources/stacks-chow/tags.txt, source lines 5852–5852. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0791: sources/stacks-chow/tags.txt, source lines 9256–9256. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 07MF. Let $X' \subset X$ and $S' \subset S$ be open subschemes such that X' maps into S'. Then there is a fully faithful functor $Cris(X'/S') \to Cris(X/S)$ which gives rise to a morphism of topoi fitting into the commutative diagram*

$$\begin{array}{ccc} (X'/S')_{cris} & \longrightarrow & (X/S)_{cris} \\ {\scriptstyle u_{X'/S'}}\downarrow & & \downarrow{\scriptstyle u_{X/S}} \\ Sh(X'_{Zar}) & \longrightarrow & Sh(X_{Zar}) \end{array}$$

Moreover, this diagram is an example of localization of morphisms of topoi as in Sites, Lemma 04H1.

Proof. The fully faithful functor comes from thinking of objects of $\mathrm{Cris}(X'/S')$ as divided power thickenings (U, T, δ) of X where $U \to X$ factors through $X' \subset X$ (since then automatically $T \to S$ will factor through S'). This functor is clearly cocontinuous hence we obtain a morphism of topoi as indicated. Let $h_{X'} \in Sh(X_{Zar})$ be the representable sheaf associated to X' viewed as an object of X_{Zar}. It is clear that $Sh(X'_{Zar})$ is the localization $Sh(X_{Zar})/h_{X'}$. On the other hand, the category $\mathrm{Cris}(X/S)/u_{X/S}^{-1}h_{X'}$ (see Sites, Lemma 0791) is canonically identified with $\mathrm{Cris}(X'/S')$ by the functor above. This finishes the proof. □

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